Computer Vision:

Week-8 (16-20 March) Theory Lectures

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Review

- Machine learning for vision
- Types of Machine learning
- Flowchart of Image classification
- Important metrics
- Numerical

Train / Validation / Test* + *Overfitting

In machine learning for computer vision*, properly dividing the dataset is essential to build reliable and generalizable models. The dataset is typically split into three parts: training, validation, and testing.*

Train / Validation / Test + *Overfitting*

The training set *is used to teach the model by allowing it to learn patterns from labeled images, such as recognizing cats, faces, or objects. The validation set acts as a tuning mechanism during training;*

Train / Validation / Test + ***Overfitting***

*it helps in adjusting **hyperparameters** like learning rate, number of layers, or model complexity without exposing the model to final evaluation data.*

Train / Validation / Test + ***Overfitting***

*Finally, the test set is used only once after training is complete to evaluate the true performance of the model on unseen data. A key issue in this process is **overfitting**, where the model performs extremely well on training data but poorly on new data because it has memorized patterns instead of learning general features.*

Train / Validation / Test + *Overfitting*

To reduce overfitting, techniques such as increasing dataset size, applying data augmentation (e.g., image rotation, flipping), and using regularization methods are commonly used.

Train / Validation / Test + *Overfitting*

Additionally, cross-validation, especially K-fold cross-validation, provides a more robust evaluation by dividing the dataset into multiple subsets and training/testing the model multiple times, ensuring that every data point is used for validation at least once.

Train / Validation / Test + *Overfitting*

To evaluate classification performance in computer vision tasks, tools like the confusion matrix, ROC curve, and Precision-Recall (PR) curve are widely used.

Train / Validation / Test + *Overfitting*

*A **confusion matrix** provides a detailed breakdown of predictions by comparing actual and predicted classes, including True Positives (correct detections), True Negatives (correct rejections), False Positives (incorrect detections), and False Negatives (missed detections)*

Train / Validation / Test + *Overfitting*

From this matrix, important metrics such as accuracy, precision, and recall are derived, where precision measures the correctness of positive predictions and recall measures the ability to detect all actual positives.

Train / Validation / Test + *Overfitting*

Why do we split data?

Train / Validation / Test + *Overfitting*

- *You teach a student using past exam questions*
- *If you test him using the same questions, he will score 100%*
- *But that does NOT mean he learned So we divide data:*

Train / Validation / Test + *Overfitting*

- ***Dataset Splitting***

Train Set

- *Used to train the model*
- *Model learns patterns here*

Example: Images of cats & dogs used to teach model

Train / Validation / Test +
Overfitting

Validation Set

- *Used to tune the model*
- *Helps choose best parameters*

Example:

- *Which model is better?*
- *How many layers?*
- *Learning rate?*

Train / Validation / Test + *Overfitting*

Test Set

- *Used only at the end*
- *Gives final performance*

Important Rule:

- *Never use test data during training*